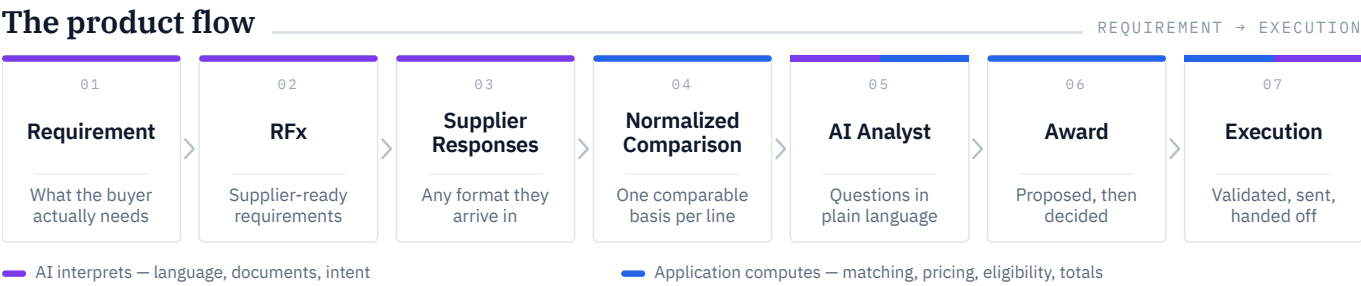


Kill the Quote Spreadsheet

AI-powered procurement from requirement to award execution

Procurement doesn't break when an RFQ is sent. It breaks when five suppliers respond differently and someone has to manually turn those responses into one spreadsheet before a decision can be made.

The system turns a buyer's requirement into an RFx, accepts supplier responses in whatever format they arrive in, extracts and normalizes the relevant information, lets the buyer interrogate the result in plain language, and carries the buyer's decision through award execution. **The model reads documents, plans queries and writes prose. Every figure on every screen is computed by the application** from a record the buyer can trace back to a supplier's own sentence.



- What we built

FIVE CAPABILITIES
- 01

Requirement Copilot

IMPLEMENTED

A vague sentence becomes a structured RFx. The model classifies the product and asks only the questions that change what a supplier would quote; every field records whether the buyer stated it or the assistant proposed it, and the two are never merged.
- 02

Supplier Response Intelligence

IMPLEMENTED

Spreadsheet, PDF, Word, plain email and a photograph of a rate card are read, matched to RFx lines and reduced to one comparable basis — with the supplier's own sentence, and its cell or page, still attached to the number.
- 03

Procurement Analyst

IMPLEMENTED

Plain-language questions answered from the same records the comparison screen renders, each answer carrying how it was calculated, which assumptions applied, and every supplier and line left out, with the reason.
- 04

Award Decision

DELIBERATELY MINIMAL

Every line is seeded with two proposals — the cheapest comparable quote, and the cheapest that also clears an explicit, buyer-set quality bar. The buyer picks per line or overrides with a reason. No weighted score: see page 3.
- 05

Award Execution

IMPLEMENTED

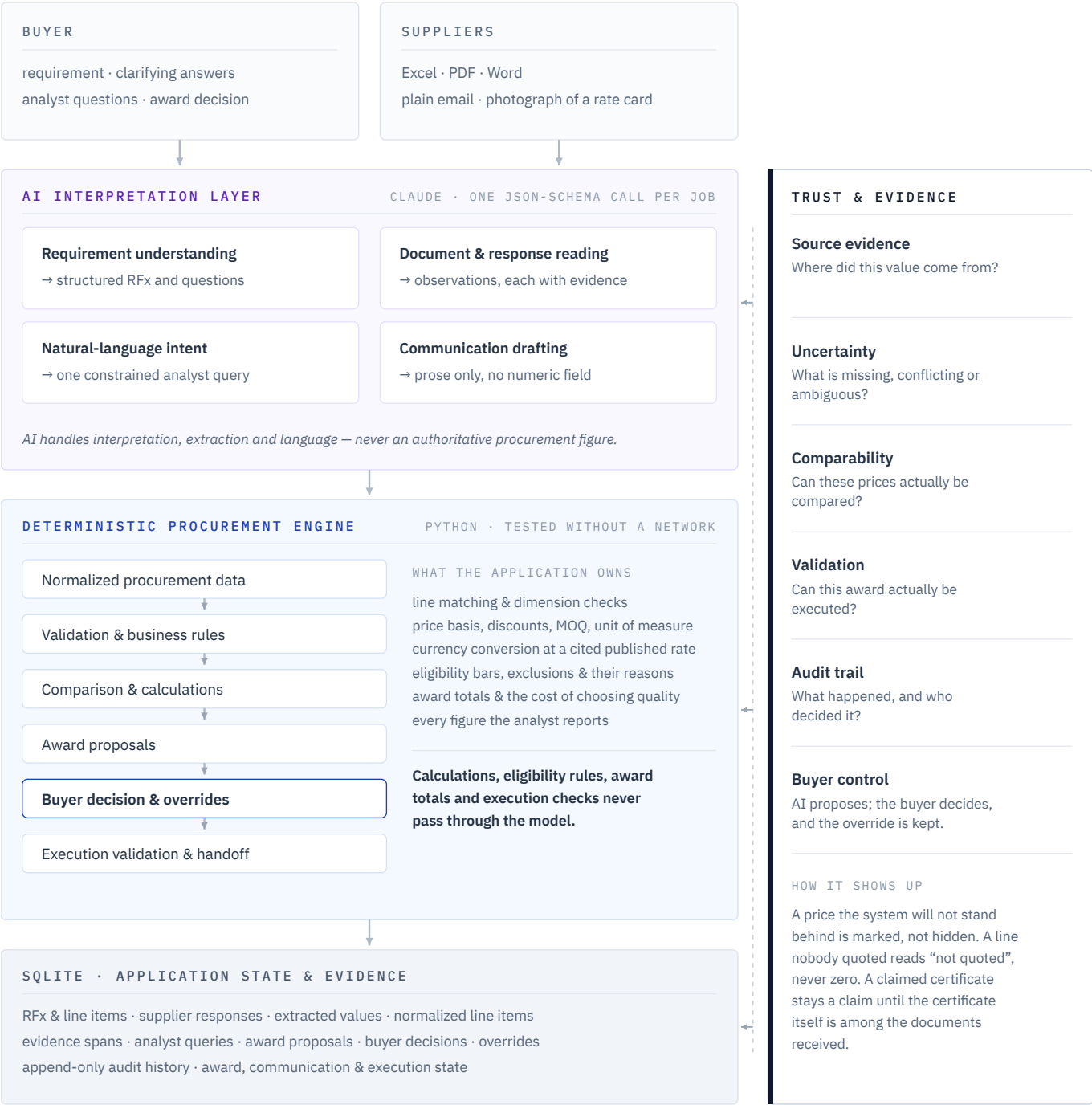
The award is validated and approved, one letter per supplier is drafted from that supplier's facts alone and checked before anyone sees it, and a structured order handoff is produced. Sending is simulated and labelled as such.

PRINCIPLE

The product isn't an AI chatbot for procurement. It is a decision system with AI at the messy interpretation points.

How the system works

AI interprets messy information; deterministic systems own procurement truth.



ARCHITECTURE PRINCIPLE

LLM ≠ Source of Truth. The model interprets unstructured information and turns natural language into structured intent. The application owns the procurement state, the calculations, the validation and the final decision.

INTERFACE	MODEL ACCESS	EVERY CALL	STATE
Streamlit	Claude Code CLI	JSON Schema validated	SQLite, append-only audit

Making it defensible

What we guaranteed, how we tested it, and what we chose not to build.

Trust by design

Evidence-backed extraction

A quoted span must be findable in the supplier's own document.

Visible uncertainty

review, *conflict*, *unresolved* and *not quoted* are distinct states.

Comparable prices only

Currencies convert only at a published rate, cited with its source.

Missing is not zero

A line nobody priced reads “not quoted”, never zero.

Claimed is not verified

A certification stays a claim until the certificate is on file.

Deterministic calculation

Every figure is computed in Python, never generated.

The buyer stays accountable

The analyst is blocked from recommending. Awarding is its own screen, and an override needs a reason.

Supplier text is data, not instruction

Instruction-shaped sentences are dropped, and the buyer is told.

Evaluation

BEYOND THE HAPPY PATH

565 automated tests across 17 files run with no network and no model calls. Six live scripts drive the real model against real binary fixtures, at up to 30 line items.

AREA	WHAT WE TESTED
Input	Vague, incomplete requirements
Supplier data	Excel, PDF, Word, email, photograph
Coverage	Lines nobody priced; a silent supplier
Commercials	Footnote discounts, MOQ, lead time
Quality	Certificate claimed vs. attached
Conflicts	A supplier contradicting itself
Revisions	A quotation superseded, original kept
Analysis	Question variants; unanswerable ones
Decision	Proposal vs. buyer override
Execution	Validation, blocking, supplier leakage

What we deliberately left out

We stubbed the external plumbing rather than spending a short prototype on integrations that prove nothing about the core thesis.

- **Email of any kind.** Sending is simulated, and labelled as simulated everywhere it appears.
- **ERP and PO integration.** The order handoff is a document you download.
- **Supplier portal, auth, approval chains.** Single-user and local.
- **Payment, invoicing, receipt, logistics.** Everything downstream of the award.
- **Weighted best-value scoring.** Replaced by an explicit, buyer-set quality bar.
- **A second model provider.** One provider, one authenticated path.

WHY

The question worth answering in a prototype was never “*can we integrate with an ERP?*” It was “*can AI turn messy supplier responses into a decision a buyer could defend?*”

What we learned

- 01 **Extraction is not the product.**
Turning documents into fields only helps if the result can carry a decision. A field with nothing behind it cannot.
- 02 **Uncertainty is a feature.**
A clean table that hides ambiguity is more dangerous than a messy one that says what to go and check.
- 03 **AI should remove work, not accountability.**
Its best role is cutting interpretation and navigation overhead while the buyer keeps every consequential decision.

What we'd build next

- Production-grade supplier communication, with clarification loops
- ERP and purchase-order integration, through to receipt
- Weighted best-value scoring on the calculation engine already in place
- Outcome tracking, so awarded performance informs the next comparison

AI should remove the spreadsheet work —
not remove the buyer from the decision.